Model training and conversion

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References

After completing the tutorial content of the dataset annotation, we can use the mainboard to start training the model.

```
This tutorial only introduces the CLI model training and conversion. You can refer to the
official website to modify the Python case yourself.
```

1. Model training

Use the CLI command to train the model directly: copy the yolo11n.pt file to the directory where the configuration file is located, and then open the terminal in the directory where the configuration file is located:

```
cd /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data
```

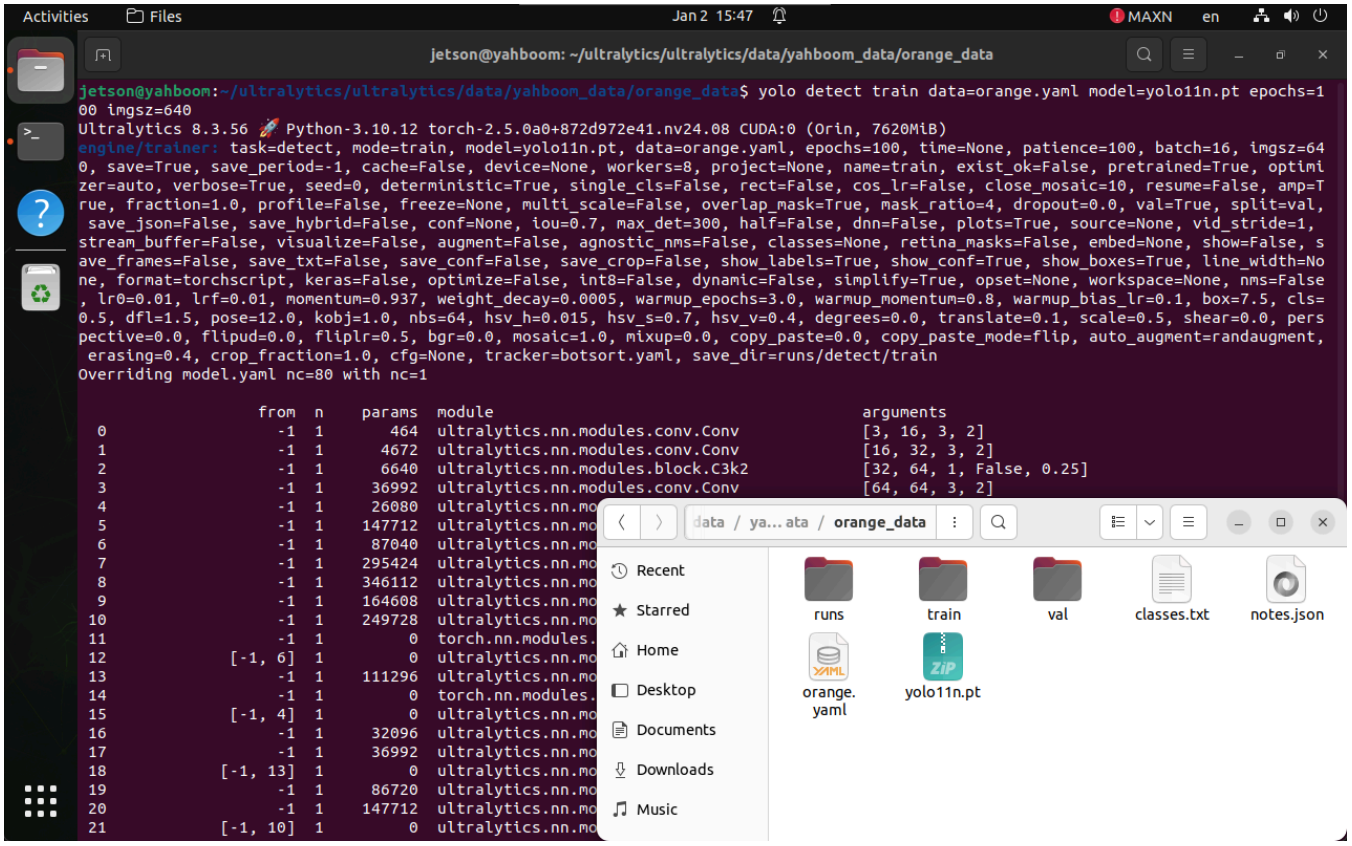
```
yolo detect train data=orange.yaml model=yolo11n.pt epochs=100 imgsz=640 amp=False
```

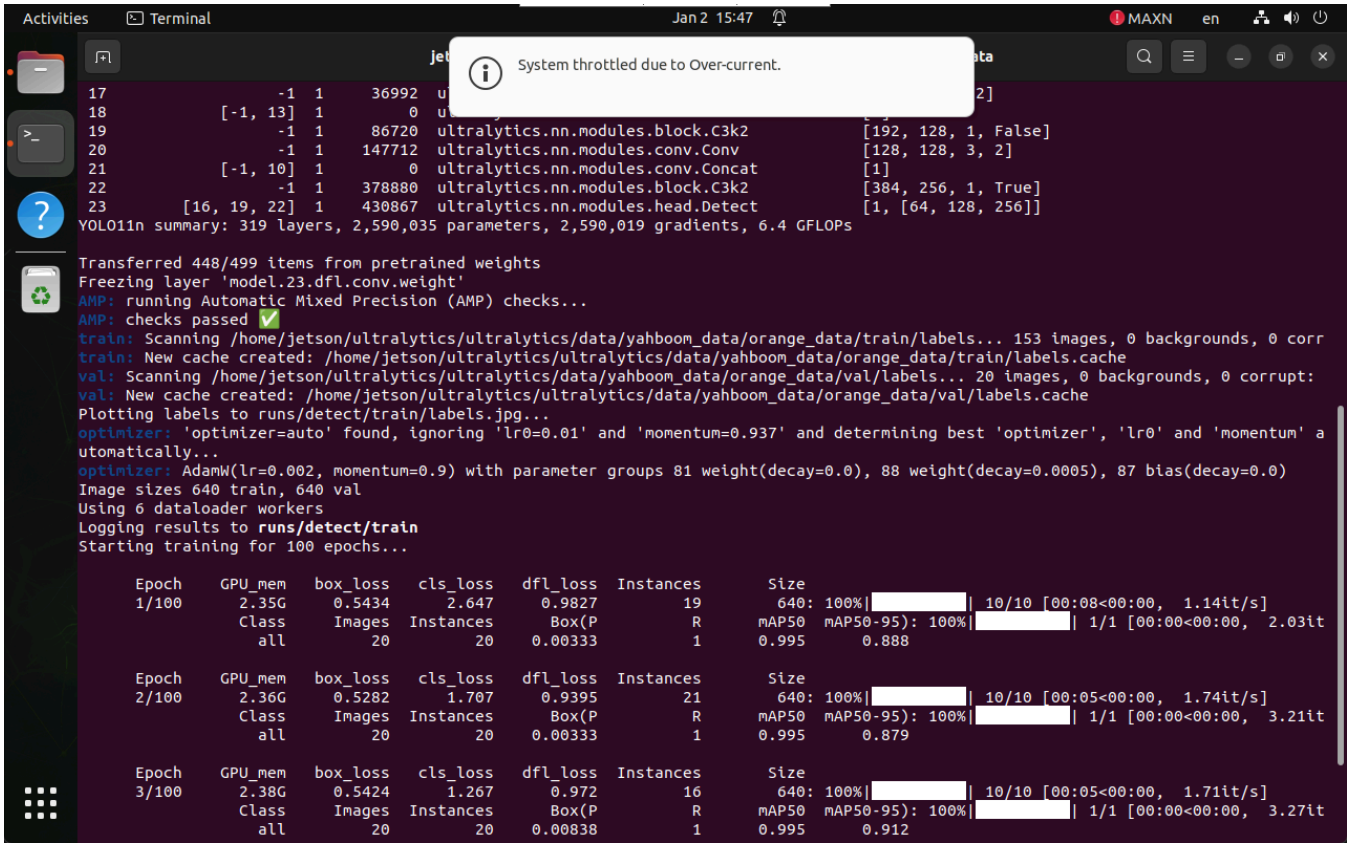
`data`: the dataset configuration file

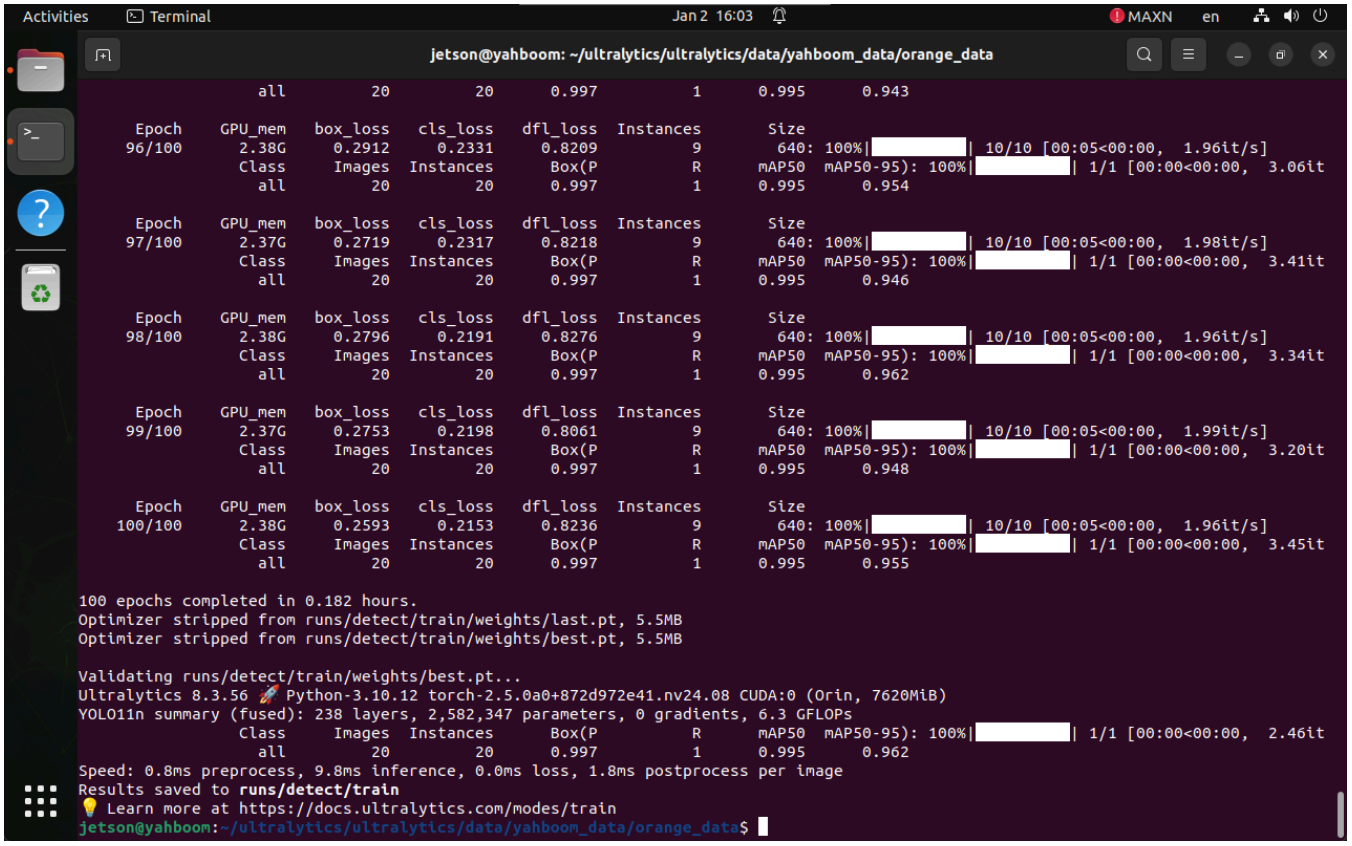
`model`: the pre-trained model file

`epochs`: the number of training epochs

`imgsz`: the specified input image size


Activities
Files
Jan 2 15:47
MAXN
en
jetson@yahboom: ~/ultralytics/ultralytics/data/yahboom_data/orange_data
jetson@yahboom:~/ultralytics/ultralytics/data/yahboom_data/orange_data$ yolo detect train data=orange.yaml model=yolo11n.pt epochs=100 imgsz=640
Ultralytics 8.3.56 🚀 Python-3.10.12 torch-2.5.0a0+872d972e41.nv24.08 CUDA:0 (Orin, 7620MiB)
engine/trainer: task=detect, mode=train, model=yolo11n.pt, data=orange.yaml, epochs=100, time=None, patience=100, batch=16, imgsz=640, save=True, save_period=-1, cache=False, device=None, workers=8, project=None, name=train, exist_ok=False, pretrained=True, optimizer=auto, verbose=True, seed=0, deterministic=True, single_cls=False, rect=False, cos_lr=False, close_mosaic=10, resume=False, amp=True, fraction=1.0, profile=False, freeze=None, multi_scale=False, overlap_mask=True, mask_ratio=4, dropout=0.0, val=True, split=val, save_json=False, save_hybrid=False, conf=None, iou=0.7, max_det=300, half=False, dnn=False, plots=True, source=None, vid_stride=1, stream_buffer=False, visualize=False, augment=False, agnostic_nms=False, classes=None, retina_masks=False, embed=None, show=False, save_frames=False, save_txt=False, save_conf=False, save_crop=False, show_labels=True, show_conf=True, show_boxes=True, line_width=None, format=torchscript, keras=False, optimize=False, int8=False, dynamic=False, simplify=True, opset=None, workspace=None, nms=False, lr0=0.01, lrf=0.01, momentum=0.937, weight_decay=0.0005, warmup_epochs=3.0, warmup_momentum=0.8, warmup_bias_lr=0.1, box=7.5, cls=0.5, dfl=1.5, pose=12.0, kobj=1.0, nbs=64, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, degrees=0.0, translate=0.1, scale=0.5, shear=0.0, perspective=0.0, flipud=0.0, fliplr=0.5, bgr=0.0, mosaic=1.0, mixup=0.0, copy_paste=0.0, copy_paste_mode=flip, auto_augment=randaugment, erasing=0.4, crop_fraction=1.0, cfg=None, tracker=botsort.yaml, save_dir=runs/detect/train
Overriding model.yaml nc=80 with nc=1
from n params module arguments
0 -1 1 464 ultralytics.nn.modules.conv.Conv [3, 16, 3, 2]
1 -1 1 4672 ultralytics.nn.modules.conv.Conv [16, 32, 3, 2]
2 -1 1 6640 ultralytics.nn.modules.block.C3k2 [32, 64, 1, False, 0.25]
3 -1 1 36992 ultralytics.nn.modules.conv.Conv [64, 64, 3, 2]
4 -1 1 26080 ultralytics.nn.mo
5 -1 1 147712 ultralytics.nn.mo
6 -1 1 87040 ultralytics.nn.mo
7 -1 1 295424 ultralytics.nn.mo
8 -1 1 346112 ultralytics.nn.mo
9 -1 1 164608 ultralytics.nn.mo
10 -1 1 249728 ultralytics.nn.mo
11 -1 1 0 torch.nn.modules.
12 [-1, 6] 1 0 ultralytics.nn.mo
13 -1 1 111296 ultralytics.nn.mo
14 -1 1 0 torch.nn.modules.
15 [-1, 4] 1 0 ultralytics.nn.mo
16 -1 1 32096 ultralytics.nn.mo
17 -1 1 36992 ultralytics.nn.mo
18 [-1, 13] 1 0 ultralytics.nn.mo
19 -1 1 86720 ultralytics.nn.mo
20 -1 1 147712 ultralytics.nn.mo
21 [-1, 10] 1 0 ultralytics.nn.mo
data / ya... ata / orange_data
Recent
Starred
Home
Desktop
Documents
Downloads
Music
runs
train
val
classes.txt
notes.json
orange.yaml
yolo11n.pt


Activities
Terminal
Jan 2 15:47
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en
System throttled due to Over-current.
17 -1 1 36992 u
18 [-1, 13] 1 0 u
19 -1 1 86720 ultralytics.nn.modules.block.C3k2 [192, 128, 1, False]
20 -1 1 147712 ultralytics.nn.modules.conv.Conv [128, 128, 3, 2]
21 [-1, 10] 1 0 ultralytics.nn.modules.conv.Concat [1]
22 -1 1 378880 ultralytics.nn.modules.block.C3k2 [384, 256, 1, True]
23 [16, 19, 22] 1 430867 ultralytics.nn.modules.head.Detect [1, [64, 128, 256]]
YOLO11n summary: 319 layers, 2,590,035 parameters, 2,590,019 gradients, 6.4 GFLOPs
Transferred 448/499 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed ✅
train: Scanning /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/train/labels... 153 images, 0 backgrounds, 0 corr
train: New cache created: /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/train/labels.cache
val: Scanning /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/val/labels... 20 images, 0 backgrounds, 0 corrupt:
val: New cache created: /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/val/labels.cache
Plotting labels to runs/detect/train/labels.jpg...
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)
Image sizes 640 train, 640 val
Using 6 dataloader workers
Logging results to runs/detect/train
Starting training for 100 epochs...
Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
1/100 2.35G 0.5434 2.647 0.9827 19 640: 100%| 10/10 [00:08<00:00, 1.14it/s]
Class Images Instances Box(P R mAP50 mAP50-95): 100%| 1/1 [00:00<00:00, 2.03it
all 20 20 0.00333 1 0.995 0.888
Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
2/100 2.36G 0.5282 1.707 0.9395 21 640: 100%| 10/10 [00:05<00:00, 1.74it/s]
Class Images Instances Box(P R mAP50 mAP50-95): 100%| 1/1 [00:00<00:00, 3.21it
all 20 20 0.00333 1 0.995 0.879
Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
3/100 2.38G 0.5424 1.267 0.972 16 640: 100%| 10/10 [00:05<00:00, 1.71it/s]
Class Images Instances Box(P R mAP50 mAP50-95): 100%| 1/1 [00:00<00:00, 3.27it
all 20 20 0.00838 1 0.995 0.912



Finally, the printed model-related files after training are stored in the runs/detect/train folder under the current directory.

Note: each time the training is run, the number after train in the path automatically increases by 1. Please record this path and use the new path of this model later.

2. Model conversion

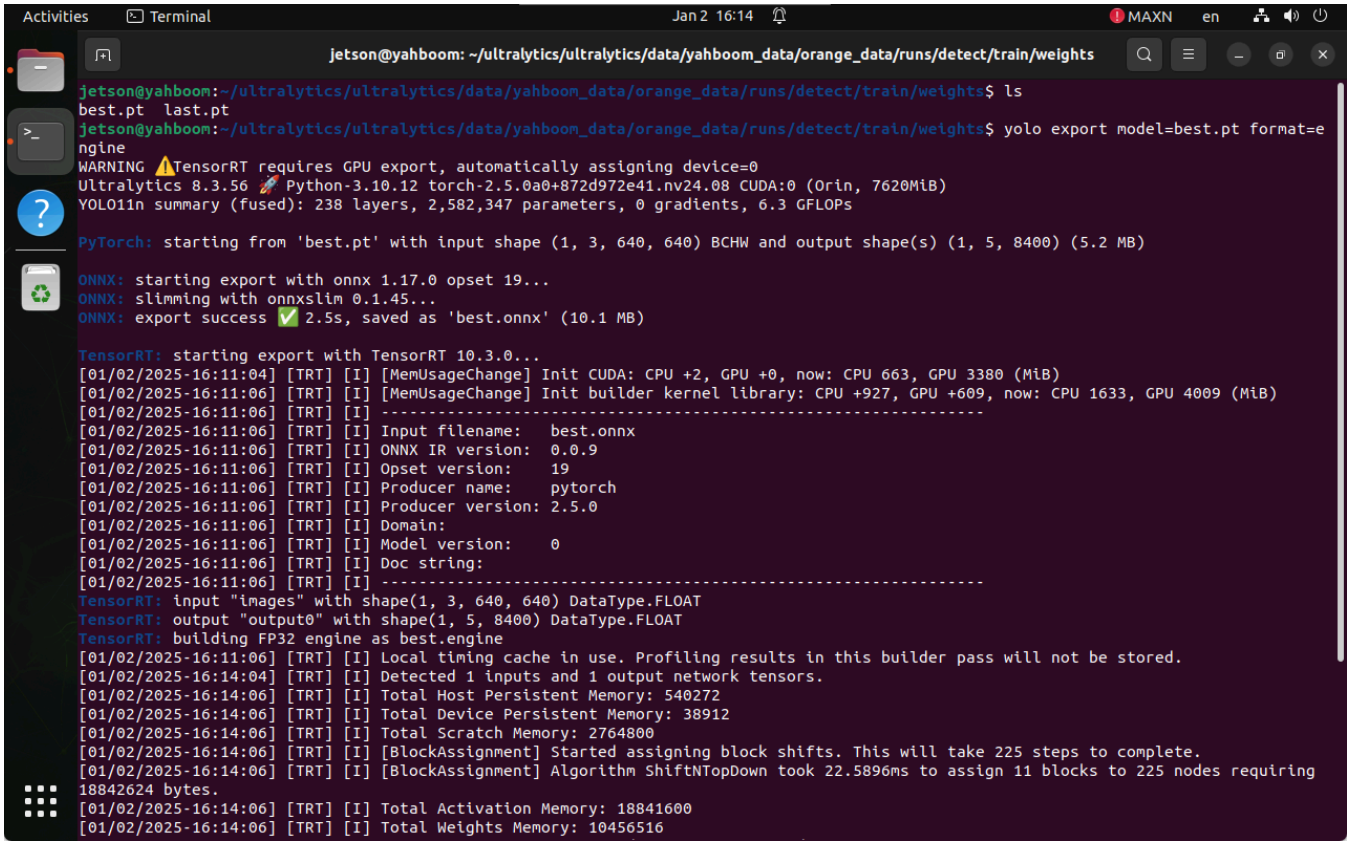
The final model is generated under the runs folder: generally the best.pt file is chosen for use

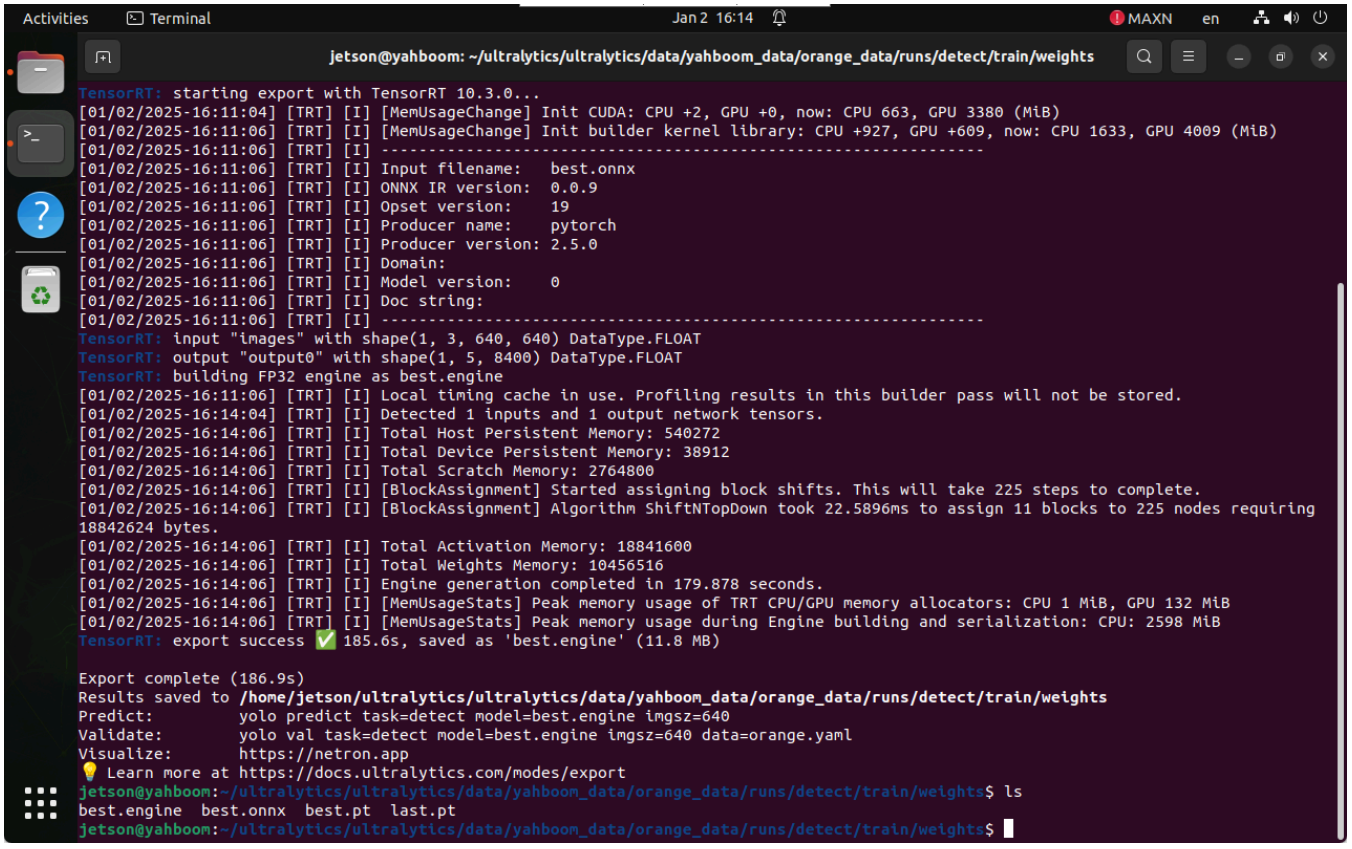
```
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights
```

Convert the PyTorch format model into TensorRT:

```
cd
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights
```

```
yolo export model=best.pt format=engine
```





References

https://docs.ultralytics.com/modes/train/